

UAS OneSpan DPMobileGateway Implementation Guide - 6.0.1-GA (AI)

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1. Preface - UAS OneSpan DPMobileGateway Implementation Guide

AccessMatrix Universal Authentication Server (UAS) supports the use of the OneSpan Digipass for Mobile Gateway (DPMobileGateway). The DPMobileGateway functions as an intermediary service between the user's mobile device and the AccessMatrix UAS server to enable secure communication and data exchange between them.

Intended Reader

This guide is intended to be a reference for the security administrator, professional service staff, and developers.

Document Scope

This document provides the concept, implementation steps and web service details of OneSpan (formerly VASCO) Digipass for Mobile Gateway (DPMobileGateway) with AccessMatrix Universal Authentication Server (UAS).

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- [UAS Administration Guide](#)
- [UAS OneSpan Multi-Device Implementation Guide](#)

2. Introduction to UAS OneSpan DPMobileGateway Implementation

UAS Digipass for Mobile Gateway (DPMobileGateway) works as an intermediary service between the mobile device (client) and AccessMatrix UAS server. It is responsible for accepting requests from the mobile device and triggers the corresponding actions by calling the AccessMatrix UAS server API to perform online activation [Secure Remote Password (SRP) / Elliptic-curve Diffie–Hellman (ECDH)] and push notification identifier registration.

This product is a stateless component written in Java without any persistence and graphical user interface.



Note:The term <AM_ROOT> will be used to refer to the root or installation directory of AccessMatrix on both Windows and Unix platforms.

Implementation Checklist

Prior to configuration, ensure that OneSpan (VASCO) VACMAN Controller token libraries are deployed properly to AccessMatrix server. Refer to the [UAS OneSpan Multi-Device Implementation Guide - Token Libraries](#) section for deployment details.

Below are the tasks that should be performed as part of the UAS OneSpan DPMobileGateway implementation checklist.

Tasks	Reference
Deploy DPMobileGateway WAR File	Deployment Steps
Configure DPMobileGateway Properties File	Configure DPMobileGateway Properties File
Enable SSL Connection	Enable SSL Connection

3. DPMobileGateway Deployment

Deployment Steps

1. Secure the DPMobileGateway WAR file from the AccessMatrix CD-Image. The filename format is **AccessMatrix-x.x.x.xxxx-GA-dp4mobile-gateway-rest.war**, e.g. **AccessMatrix-5.6.7.6711-dp4mobile-gateway-rest.war**).
2. (Optional) For easier reference, rename the WAR file to **dpmgateway.war**.
3. DPMobileGateway is a stateless web service, therefore you can deploy the WAR file (**dpmgateway.war**) to any web application container supported by AccessMatrix core products, e.g. Apache Tomcat.
4. If using Tomcat, create a *dpmgateway* folder under *<AM_ROOT>\tomcat\webapps*.
5. Extract the content of **dpmgateway.war** to this folder.

DPMobileGateway Components

Below is the list of the folders and sub-folders inside the DPMobileGateway WAR file:

Folder/File Name	Sub-folder/File Name	Description
META-INF	MANIFEST.MF	Contains the manifest version of the WAR file.
WEB-INF	classes\log4j.properties	Contains the configurations of log level and log output file location.
	classes\log4j.xml	Contains the configurations of log level and log output file location.
	conf\config.properties	Contains the configurations of connection to AccessMatrix server, proxy name, and digipass for mobile token.
	conf\testagent.keystore	An agent keystore used for testing.
	lib	The libraries used for DPMobileGateway.
	web.xml	Contains filter mapping and servlet mapping.
index.html	-	The web application's default page.

4. DPMobileGateway Configuration

This section defines the steps relating to the configuration of the UAS DPMobileGateway solution.

- **Configure DPMobileGateway Properties File**

Before DPMobileGateway can be used with AccessMatrix UAS server, DPMobileGateway properties file must be properly configured. Refer to [Configure DPMobileGateway Properties File](#) for configuration details.

- **Enable SSL Connection**

SSL connection must be enabled to keep sensitive information sent across the internet encrypted and secured. Refer to [Enable SSL Connection](#) for configuration details.

- **Verify DPMobileGateway Connection**

Once AccessMatrix UAS server start up is successful, launch any browser and enter the DPMobileGateway URL: *https://<server_host>:<server_port>/dpmgateway*, e.g.

https://localhost:8443/dpmgateway. If connection is successful, the "Welcome to Digipass for Mobile Gateway" message will be displayed.

Configure DPMobileGateway Properties File & DIGIPASS.xml

To configure DPMobileGateway properties file, navigate to

`<AM_ROOT>\tomcat\webapps\dpmgateway\WEB-INF\conf\` and modify the **config.properties** file. The content is shown below:

Properties (.properties files)

```
#Language
locale.1=en

# PART 1 - REST Configuration
HOSTS=http://localhost:8080/am5/rest
#TRUST_STORE=C:/apache-tomcat-7.0.57/webapps/DP4Mobile/WEB-INF/classes/testtrust.keystore
#AGENT_KEYSTORE=C:/apache-tomcat-7.0.57/webapps/DP4Mobile/WEB-INF/classes/testagent.keystore
#KEYSTORE_PWD=passphrase
ConnTimeoutSecs=60

# PART 2 - Login Admin Configuration
ADMIN_LOGIN_REALMID=40153
ADMIN_LOGIN_ID=usoagent
ADMIN_PASSWORD=
USER_STORE_ID=DefaultStore
```

```
# PART 3- DigipassForMobile Token Configuration
DPM_1FA_REALMID=40151
DPM_USER_STORE_ID=
DPM_TOKEN_MODEL=DAL10
DPM_TOKEN_GROUP=DPMobileGroup1
DPM_TOKEN_STORE=
DPM_TOKEN_VENDOR=VASCO_MULTI_DEVICE
DPM_AUTH_MODE=RO

# PART 4 - Activation Password/Authorization Code Feature Configuration

# Online Activation Protocol
DPM_ONLINE_ACTIVATION_SRP_OR_ECDH=ECDH

#DigipassForMobile useActivationPassword - this value must be in sync
with the customization file setting e.g.
# <ActivationData
#   . .
#   useActivationPassword="true"
# .. >
# If set to "true", please set DPM_ONLINE_ACTIVATION_SRP_OR_ECDH to
"ECDH"
DPM_ACTIVATION_WITH_ACTIVATION_PASSWORD=true

#Start of Online Activation
# Set this to send Activation Password through event notification. It is
mandatory now for current gateway calling AM REST API.
# If not set, the Activation Password will be embedded in the activation
CrontoSign/QRCode image
# can use default created policy - DPMobileSMSActivationPasswordPolicy
DPM_EVENT_NOTIFICATION_POLICY_ID_ONLINE=DPMobileSMSActivationPasswordPol
icy

#DigipassForMobile useAuthorizationCode - this value must be in sync with
the customization file setting e.g.
# <OnlineActivationData
#   . .
#   useAuthorizationCode="true"
# .. >
DPM_ONLINE_ACTIVATION_WITH_AUTHORIZATION_CODE= true
# Set this to send Authorization Code through event notification. It is
mandatory now for gateway calling AM REST API.
# If not set, the authorization code will be embedded in the activation
CrontoSign/QRCode image
# can use default created policy - DPMobileEmailAuthorizationCodePolicy
DPM_EVENT_NOTIFICATION_POLICY_ID_ONLINE_AUTH_CODE=DPMobileEmailAuthorizat
ionCodePolicy
#End of Online Activation

#Start of Offline Activation
# If DPM_ACTIVATION_WITH_ACTIVATION_PASSWORD= true,
DPM_EVENT_NOTIFICATION_POLICY_ID_OFFLINE must be set
DPM_EVENT_NOTIFICATION_POLICY_ID_OFFLINE=DPMobileSMSActivationPasswordPol
icy
#End of Offline Activation

#DigipassForMobile post activation
```

```

DPM_ONLINE_POST_ACTIVATION_WITH_USER_IDENTIFIER=true
DPM_SERVER_ACTIVATION_CR =true

#DPMobile Success/Fail Page
DPM_SERVLET_SUCCESS_URL=DigipassMobileRegSuccess.jsp
DPM_SERVLET_FAIL_URL=DigipassMobileRegFail.jsp

# PART 5 - DPM Download link
DPM_DOWNLOAD_LINK_ONLINE=http://hostname:port/dpmobile/download_online.js
p
DPM_DOWNLOAD_LINK_OFFLINE=http://hostname:port/dpmobile/download_offline.
jsp
#Path for DPMobile APPS - on AppStore (Strictly for Apple's device),
PlayStore, or on private hosted server (only for Android. Blackberry and
Javaphone)
DPM_IOS_PATH_ONLINE=http://itunes.apple.com/<country>/app/<app\u2013name>
/id<app-ID>?mt=8
DPM_IOS_PATH_OFFLINE=http://itunes.apple.com/<country>/app/<app\u2013name>
/id<app-ID>?mt=8
DPM_ANDROID_PATH_ONLINE=android/DIGIPASS_online.apk
DPM_ANDROID_PATH_OFFLINE=android/DIGIPASS_offline.apk
DPM_JAVAPHONE_PATH=midp/DIGIPASS.jad
DPM_BLACKBERRY_PATH=blackberry/ota/DIGIPASS.jad

#Notification Setting
# action identifier must be the same as the one configure in DIGIPASS.xml
DPM_NOTIFICATION_ACTION_IDENTIFIER=notificationTds
# connection timeout in milliseconds
DPM_NOTIFICATION_SETTING_CONN_TIMEOUT=20000
# terminate timeout in milliseconds
DPM_NOTIFICATION_SETTING_TERMINATE_TIMEOUT=60000
# max number of threads allowed to run for sending push messages
DPM_NOTIFICATION_SETTING_THREAD_MAX_NUMBER=20

# PART 6 - Notification Message Construct
DPM_NOTIFICATION_ACTION_IDENTIFIER=notificationTds
DPM_NOTIFICATION_TITLE=Pending Transaction
DPM_NOTIFICATION_SUBJECT=There are pending transaction(s) waiting for
your approval
DPM_NOTIFICATION_VERSION_TAG=01
DPM_PUSH_LOGIN_NOTIFICATION_ACTION_IDENTIFIER=notificationLogin
DPM_PUSH_LOGIN_NOTIFICATION_TITLE=Push Login
DPM_PUSH_LOGIN_NOTIFICATION_SUBJECT=Login request pending for your
approval

# PART 7 - SQLite DB for Transaction/Session Storing - for Scan & Login
and Push & Sign feature
# If using Push2Login feature, change
DPM_APPLICATION_SAMPLE_DATABASE_FULLPATH=AM
DPM_APPLICATION_SAMPLE_DATABASE_FULLPATH=C:\\database\\dp4mobile-ws.db
# Login Module Id must be set for Push2Login feature
#DPM_PUSH2LOGIN_LOGIN_MODULE_ID=Push2Login
#Scan & Sign session id length
DPM_SESSION_ID_LENGTH=6

```

The table below explains the important properties to configure in **config.properties** file:

Property Name	Description	Remarks
HOSTS	This property refers to the AccessMatrix server REST API endpoint URL, e.g. https://<server_host>:<server_port>/am5/rest	-
ConnTimeoutSecs	This property refers to the connection timeout for API call. Default: 60 seconds.	-
DPM_TOKEN_VENDOR	This property refers to the token vendor type to be used. Default: VASCO_MULTI_DEVICE	Only supports VASCO Multi-Device token types (i.e., VASCO_MULTI_DEVICE, VASCO_MULTI_DEVICE_FM_S AFENET, and VASCO_MULTI_DEVICE_FM_T HALES).
DPM_ACTIVATION_SECURE_PROTOCOL	This property refers to the online activation protocol to be used. Options: ECDH or SRP Default: ECDH	Make sure the protocol configured here corresponds to the Secured Data Exchange Protocol value configured in Administration Dashboard > Security Policy > Token > Token Policy > DefaultVascoKernelParameters > Digipass for APPS/Mobile Specific tab.

Save the changes.

Add deviceFriendlyName to DIGIPASS.xml

Add the deviceFriendlyName payload parameter in the **DIGIPASS.xml** file as shown below (replace <Your Device Friendly Name> with the actual device name):

XML

```
<InstanceActivation>
  <OnlineInstanceActivation >
    <URL method="POST"
value="http://192.168.1.98:8085/dpmobile/DPMobileGateway">
      <PayloadParameter key="action"
value="instanceActivation" />
```

```
        <PayloadParameter key="serialNumber"
value="%_SerialNumber_%" />
        <PayloadParameter key="deviceCode"
value="%_DeviceCode_%" />
        *<PayloadParameter
key="deviceFriendlyName" value="<Your Device Friendly
Name>" />*
    </URL>
</OnlineInstanceActivation>
```

Enable SSL Connection

The following steps will set up the SSL connection between the Digipass mobile gateway and the AccessMatrix UAS server:

1. Navigate to `<AM_ROOT>\tomcat\conf\` and modify the **server.xml** file.
 2. Under SSL Connector (e.g. 8443 port), set the `clientAuth` to `false`.
 3. Save the changes.
 4. Start AccessMatrix UAS server.
-

5. DPMobileGateway Implementation using API

DPMobile Gateway Use Cases

The primary use cases of UAS DPMobile Gateway are:

- [Registration Request](#)
- [License Activation for SRP](#)
- [License Activation for ECDH](#)
- [Instance Activation](#)
- [Post Activation](#)
- [Notification Registration](#)
- [Time Synchronization](#)

5.1. Registration Request

- This service is for SRP online activation to initiate a registration request to the server.
- The objective is to trigger SRP Server Ephemeral Keys generation by AccessMatrix UAS server.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "registrationRequest".	Yes
userIdentity	String	DPMobileGateway expects this parameter as a token serial number.	Yes
authorizationCode	String	Authorization code generated by AccessMatrix UAS server.	No

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	DPMobileGateway return message (success or error message).
salt	String	AccessMatrix UAS server-generated random salt.
serverEphemeralPublicKey	String	AccessMatrix UAS server-generated "Server Ephemeral Public Key" for SRP.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httppost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action",
"registrationRequest"));
    nameValuePairs.add(new BasicNameValuePair("userIdentity", "VDS1000120"));
    nameValuePairs.add(new BasicNameValuePair("authorizationCode",
"AbCdEf123456"));
    httppost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

    // Execute HTTP Post Request
    HttpResponse postResponse = httpClient.execute(httppost);
    String responseBody = EntityUtils.toString(postResponse.getEntity());
    response = parseSerializedData(responseBody);

    String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
    String message = response.get("message"); //return message from server.
e.g. success/error message
    String serverEphemeralPublicKey =
response.get("serverEphemeralPublicKey");
    String salt = response.get("salt");

    /**
     * Continue processing the above returned values
     * with VASCO client-side SDK (e.g. DigipassSDK)
     */

} catch (ClientProtocolException e) {
    // TODO Auto-generated catch block
} catch (IOException e) {
    // TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
```

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
Map<String, String> response = new HashMap<String, String>();
String[] keyValues = data.split(";");
for (String keyValue : keyValues)
{
    String[] tmp = keyValue.split("=");
    response.put(tmp[0], tmp[1]);
}
return response;
}
```

Sample HTTP POST Request and Response

The screenshot shows the 'Variables' window in Android Studio, displaying the state of variables during an HTTP POST request. The variables are as follows:

- static members of Utils**:
 - action** = "registrationRequest"
 - content** = "authorizationCode=783805;userIdentity=VDS1000120"
 - multiDevice** = true
- dpmgatewayUrl** = "http://192.168.50.50:8081/dp4mobile/DPMobileGateway"
- httpClient** = {DefaultHttpClient@6113}
- httpPost** = {HttpPost@6114}
- mapContent** = {HashMap@6115} size = 2
- paramValuePair** = {ArrayList@6116} size = 3
 - 0 = {BasicNameValuePair@6129} "action=registrationRequest"
 - 1 = {BasicNameValuePair@6130} "authorizationCode=783805"
 - 2 = {BasicNameValuePair@6131} "userIdentity=VDS1000120"
- postResponse** = {BasicHttpResponse@6117}
- responseBody** = "salt=1A618162106D0BAA5DE44247F2A59F38;retCode=0;serverEphemeralPublicKey=2F7F1D7DB7C1D338DAFD334C610CF4C1EC7CB50E03263A0A381F663AE5B1... View"
- response** = {HashMap@6127} size = 3
 - 0 = {HashMap\$Node@6137} "salt" -> "1A618162106D0BAA5DE44247F2A59F38"
 - 1 = {HashMap\$Node@6138} "retCode" -> "0"
 - 2 = {HashMap\$Node@6139} "serverEphemeralPublicKey" -> "2F7F1D7DB7C1D338DAFD334C610CF4C1EC7CB50E03263A0A381F663AE5B1DA0678FB467056AC7A692B665B671F9CD44245183... View"

5.2. License Activation for SRP

- This service is to be called after [Registration Request](#).
- This service is meant to start the License Activation process for SRP online activation.
- The objective is to trigger the License Activation process by AccessMatrix UAS server which in return will generate license activation message.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "licenseActivation".	Yes
userIdentity	String	DPMobileGateway expects this parameter as a token serial number.	Yes
clientEphemeralPublicKey	String	Client Ephemeral Public Key generated by the Digipass APP for SRP online activation.	Yes
clientEvidenceMessage	String	Client Evidence Message generated by Digipass APP for SRP online activation.	Yes
deviceIdentifier	String	Device unique identifier. If this parameter is passed, AccessMatrix UAS server will perform device uniqueness check.	No
authorizationCode	String	Authorization code generated by AccessMatrix UAS server.	No

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	DPMobileGateway return message (success or error message).
encryptedLicenseActivationMessage	String	UAS server-generated encrypted license activation message.
encryptedLicenseActivationMessageMAC	String	UAS server-generated encrypted license activation message for MAC.
encryptedLicenseActivationMessageEncryptionCounter	String	UAS server-generated encrypted license activation message encryption counter.
serverEvidenceMessage	String	UAS server-generated server evidence message for SRP.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway	
Java	<pre>HttpClient httpClient = new DefaultHttpClient(); Map<String, String> response = new HashMap<String, String>(); HttpPost httppost = new HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway"); try { // Add your data List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2); nameValuePairs.add(new BasicNameValuePair("action", "licenseActivation")); nameValuePairs.add(new BasicNameValuePair("userIdentity", "VDS1000120")); nameValuePairs.add(new BasicNameValuePair("authorizationCode", "AbCdEf123456")); }</pre>

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
nameValuePairs.add(new BasicNameValuePair("clientEphemeralPublicKey",
"AbCdEf123456"));
nameValuePairs.add(new BasicNameValuePair("clientEvidenceMessage",
"AbCdEf123456"));
nameValuePairs.add(new BasicNameValuePair("deviceIdentifier",
"AbCdEf123456"));
httppost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

// Execute HTTP Post Request
HttpResponse postResponse = httpClient.execute(httppost);
String responseBody = EntityUtils.toString(postResponse.getEntity());
response = parseSerializedData(responseBody);

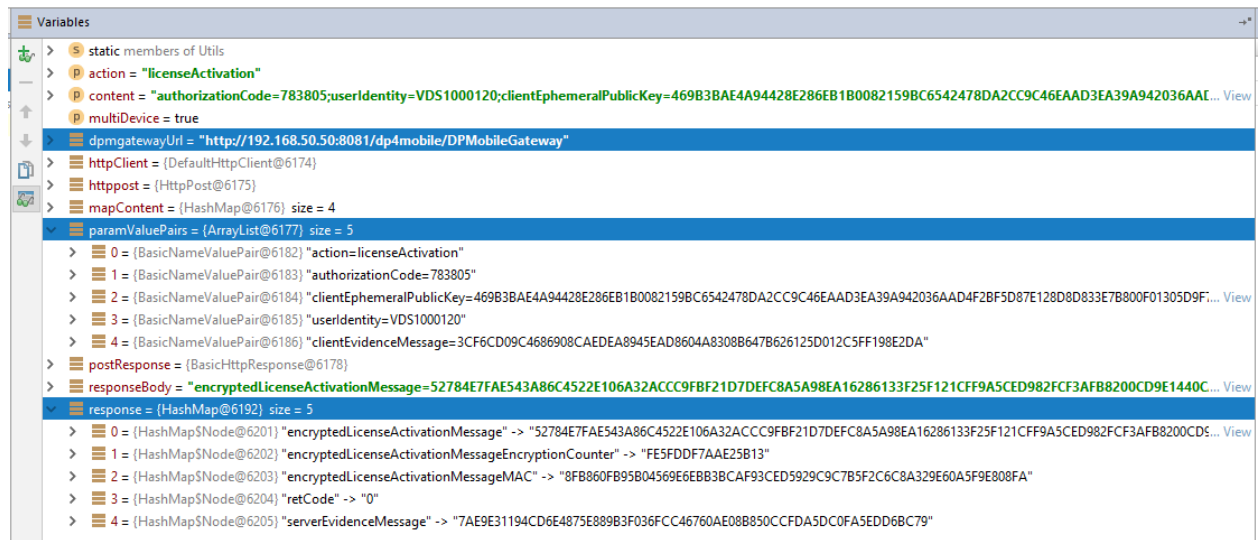
String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
String message = response.get("message"); //return message from server.
e.g. success/error message
String encryptedLicenseActivationMessage =
response.get("encryptedLicenseActivationMessage");
String encryptedLicenseActivationMessageMAC =
response.get("encryptedLicenseActivationMessageMAC");
String encryptedLicenseActivationMessageEncryptionCounter =
response.get("encryptedLicenseActivationMessageEncryptionCounter");
String serverEvidenceMessage = response.get("serverEvidenceMessage");

/**
 * Continue processing the above returned values
 * with VASCO client-side SDK (e.g. DigipassSDK)
 */

} catch (ClientProtocolException e) {
// TODO Auto-generated catch block
} catch (IOException e) {
// TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
Map<String, String> response = new HashMap<String, String>();
String[] keyValues = data.split(";");
for (String keyValue : keyValues)
{
String[] tmp = keyValue.split("=");
response.put(tmp[0], tmp[1]);
}
return response;
}
```

Sample HTTP POST Request and Response



The screenshot displays the 'Variables' tab in a Java IDE, showing the state of an HTTP client and its response. The variables are organized into a tree structure on the left, with the main details in the right pane.

Variables:

- static members of Utils**
- action = "licenseActivation"**
- content = "authorizationCode=783805;userIdentity=VDS1000120;clientEphemeralPublicKey=469B3BAE4A94428E286EB1B0082159BC6542478DA2CC9C46EAD3EA39A942036AAE..."**
- multiDevice = true**
- dpmgatewayUrl = "http://192.168.50.50:8081/dp4mobile/DPMobileGateway"**
- httpClient = {DefaultHttpClient@6174}**
- httpPost = {HttpPost@6175}**
- mapContent = {HashMap@6176} size = 4**
- paramValuePairs = {ArrayList@6177} size = 5**
 - 0 = {BasicNameValuePair@6182} "action=licenseActivation"
 - 1 = {BasicNameValuePair@6183} "authorizationCode=783805"
 - 2 = {BasicNameValuePair@6184} "clientEphemeralPublicKey=469B3BAE4A94428E286EB1B0082159BC6542478DA2CC9C46EAD3EA39A942036AAE..."
 - 3 = {BasicNameValuePair@6185} "userIdentity=VDS1000120"
 - 4 = {BasicNameValuePair@6186} "clientEvidenceMessage=3CF6CD09C4686908CAEDEA8945EAD8604A8308B647B626125D012C5FF198E2DA"
- postResponse = {BasicHttpResponse@6178}**
- responseBody = "encryptedLicenseActivationMessage=52784E7FAE543A86C4522E106A32ACCC9F8F21D7DEF8A5A98EA16286133F25F121CFF9A5CED982FCF3AFB8200CD9E1440C..."**
- response = {HashMap@6192} size = 5**
 - 0 = {HashMap\$Node@6201} "encryptedLicenseActivationMessage" -> "52784E7FAE543A86C4522E106A32ACCC9F8F21D7DEF8A5A98EA16286133F25F121CFF9A5CED982FCF3AFB8200CD9E1440C..."
 - 1 = {HashMap\$Node@6202} "encryptedLicenseActivationMessageEncryptionCounter" -> "FE5FDDF7AAE25B13"
 - 2 = {HashMap\$Node@6203} "encryptedLicenseActivationMessageMAC" -> "8FB860FB95B04569E6EBB3BCAF93CED5929C9C7B5F2C6C8A329E60A5F9E808FA"
 - 3 = {HashMap\$Node@6204} "retCode" -> "0"
 - 4 = {HashMap\$Node@6205} "serverEvidenceMessage" -> "7AE9E31194CD6E4875E889B3F036FCC46760AE08B850CCFDA5DC0FA5EDD6B8C79"

5.3. License Activation for ECDH Protocol

i **Note:** For ECDH online activation, the [Registration Request](#) step is **NOT** required. You may proceed directly to license activation.

- This service is meant to start the License Activation process for ECDH online activation.
- The objective is to trigger License Activation process by AccessMatrix UAS server which in return will generate license activation message.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway

i **Note:** <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "licenseActivation".	Yes
registrationIdentifier	String	DPMobileGateway expects this parameter as a token serial number.	Yes
publicKey	String	Client Public Key generated by the Digipass APP for ECDH online activation.	Yes
initialVector	String	Client's random Initial Vector generated by the Digipass APP for ECDH online activation.	Yes
deviceIdIdentifier	String	Device unique identifier. If this parameter is passed, AccessMatrix UAS server will perform device uniqueness checking.	No
authorizationCode	String	Authorization code generated by AccessMatrix UAS server.	No

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	DPMobileGateway return message (success or error message).
encryptedLicenseActivationMessage	String	UAS server-generated encrypted license activation message.
licenseActivationMessageIV	String	UAS server-generated license activation message initial vector.
encryptedServerPublicKey	String	UAS server-generated encrypted server public key.
encryptedNonces	String	UAS server-generated encrypted nonces.
generateSessionKeyIV	String	UAS server-generated initial vector for session key generation.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httppost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action", "licenseActivation"));
    nameValuePairs.add(new BasicNameValuePair("registrationIdentifier",
"VDS1000120"));
    nameValuePairs.add(new BasicNameValuePair("authorizationCode",
"AbCdEf123456"));
    nameValuePairs.add(new BasicNameValuePair("publicKey", "AbCdEf123456"));
    nameValuePairs.add(new BasicNameValuePair("initialVector",
"AbCdEf123456"));
}
```

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
nameValuePairs.add(new BasicNameValuePair("deviceIdentifier",
"AbCdEf123456"));
httppost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

// Execute HTTP Post Request
HttpResponse postResponse = httpClient.execute(httppost);
String responseBody = EntityUtils.toString(postResponse.getEntity());
response = parseSerializedData(responseBody);

String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
String message = response.get("message"); //return message from server.
e.g. success/error message
String encryptedLicenseActivationMessage =
response.get("encryptedLicenseActivationMessage");
String licenseActivationMessageIV =
response.get("licenseActivationMessageIV");
String encryptedServerPublicKey =
response.get("encryptedServerPublicKey");
String encryptedNonces = response.get("encryptedNonces");
String generateSessionKeyIV = response.get("generateSessionKeyIV");

/**
 * Continue processing the above returned values
 * with VASCO client-side SDK (e.g. DigipassSDK)
 */

} catch (ClientProtocolException e) {
// TODO Auto-generated catch block
} catch (IOException e) {
// TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
Map<String, String> response = new HashMap<String, String>();
String[] keyValues = data.split(";");
for (String keyValue : keyValues)
{
String[] tmp = keyValue.split("=");
response.put(tmp[0], tmp[1]);
}
return response;
}
```

5.4. Instance Activation

- This service is for online activation to initiate an Instance Activation request to the server.
- The objective is to trigger Instance Activation Message generation by UAS server.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "instanceActivation".	Yes
serialNumber	String	DPMobileGateway expects this parameter as a token serial number.	Yes
deviceCode	String	The device code generated by Digipass APP after license activation process.	Yes

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	DPMobileGateway return message (success or error message).
instanceActivationMessage	String	UAS server-generated "Instance Activation Message". This return value is the encrypted token secret.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httppost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action",
"instanceActivation"));
    nameValuePairs.add(new BasicNameValuePair("serialNumber", "VDS1000120"));
    nameValuePairs.add(new BasicNameValuePair("deviceCode", "AbCdEf123456"));
    httppost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

    // Execute HTTP Post Request
    HttpResponse postResponse = httpClient.execute(httppost);
    String responseBody = EntityUtils.toString(postResponse.getEntity());
    response = parseSerializedData(responseBody);

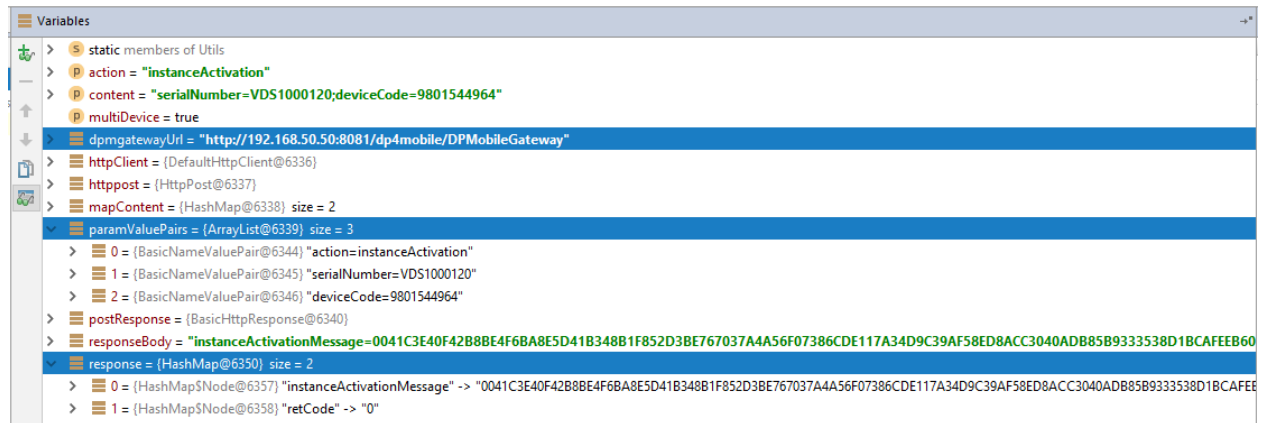
    String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
    String message = response.get("message"); //return message from server.
e.g. success/error message
    String instanceActivationMessage =
response.get("instanceActivationMessage");

    /**
     * Continue processing the above returned values
     * with VASCO client-side SDK (e.g. DigipassSDK)
     */

} catch (ClientProtocolException e) {
    // TODO Auto-generated catch block
} catch (IOException e) {
    // TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
    Map<String, String> response = new HashMap<String, String>();
    String[] keyValues = data.split(";");
    for (String keyValue : keyValues)
    {
        String[] tmp = keyValue.split("=");
        response.put(tmp[0], tmp[1]);
    }
    return response;
}
```

Sample HTTP POST Request and Response



```
Variables
> static members of Utils
> action = "instanceActivation"
> content = "serialNumber=VDS1000120;deviceCode=9801544964"
> multiDevice = true
> dpmgatewayUrl = "http://192.168.50.50:8081/dp4mobile/DPMobileGateway"
> httpClient = {DefaultHttpClient@6336}
> httpPost = {HttpPost@6337}
> mapContent = {HashMap@6338} size = 2
> paramValuePairs = {ArrayList@6339} size = 3
  > 0 = {BasicNameValuePair@6344} "action=instanceActivation"
  > 1 = {BasicNameValuePair@6345} "serialNumber=VDS1000120"
  > 2 = {BasicNameValuePair@6346} "deviceCode=9801544964"
> postResponse = {BasicHttpResponse@6340}
> responseBody = "instanceActivationMessage=0041C3E40F42B8BE4F6BA8E5D41B348B1F852D3BE767037A4A56F07386CDE117A34D9C39AF58ED8ACC3040ADB85B9333538D1BCAFEEB60"
> response = {HashMap@6350} size = 2
  > 0 = {HashMap$Node@6357} "instanceActivationMessage" -> "0041C3E40F42B8BE4F6BA8E5D41B348B1F852D3BE767037A4A56F07386CDE117A34D9C39AF58ED8ACC3040ADB85B9333538D1BCAFEEB60"
  > 1 = {HashMap$Node@6358} "retCode" -> "0"
```

5.5. Post Activation

- This service is for online activation to initiate Post Activation request to the server.
- The objective is to verify the activation and subsequently allows UAS server to update the activated token instance status to Activated.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "postActivation".	Yes
serialNumber	String	DPMobileGateway expects this parameter as a token serial number.	Yes
sequenceNumber	String	Sequence number of the activated instance (obtain through Digipass SDK) Value: 1,2 3, ,99	Yes
otp	String	Response OTP generated by the Digipass based on the instance activation message.	Yes

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	A string value of "You have been successfully authenticated." will be returned if the post-activation is successful.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httpPost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action", "postActivation"));
    nameValuePairs.add(new BasicNameValuePair("serialNumber", "VDS1000120"));
    nameValuePairs.add(new BasicNameValuePair("sequenceNumber", "1"));
    nameValuePairs.add(new BasicNameValuePair("otp", "12345678"));
    httpPost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

    // Execute HTTP Post Request
    HttpResponse postResponse = httpClient.execute(httpPost);
    String responseBody = EntityUtils.toString(postResponse.getEntity());
    response = parseSerializedData(responseBody);

    String retCode = response.get("retCode"); //return code, 0 means
    successful, failed if otherwise
    String message = response.get("message"); //return message from server.
    e.g. success/error message

    /**
     * Continue processing the above returned values
     * with VASCO client-side SDK (e.g. DigipassSDK)
     */

} catch (ClientProtocolException e) {
    // TODO Auto-generated catch block
} catch (IOException e) {
    // TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
    Map<String, String> response = new HashMap<String, String>();
    String[] keyValues = data.split(";");
    for (String keyValue : keyValues)
    {
```

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
String[] tmp = keyValue.split("=");  
response.put(tmp[0], tmp[1]);  
}  
return response;  
}
```

Sample HTTP POST Request and Response

The screenshot shows the 'Variables' window in an IDE, displaying the state of variables during an HTTP POST request. The variables are organized into a tree structure with expandable nodes.

- static members of Utils**
 - action** = "postActivation"
 - content** = "serialNumber=VDS1000120;otp=1487006;sequenceNumber=7"
 - multiDevice** = true
- dpmgatewayUrl** = "http://192.168.50.50:8081/dp4mobile/DPMobileGateway"
- httpClient** = {DefaultHttpClient@6374}
- httpPost** = {HttpPost@6375}
- mapContent** = {HashMap@6376} size = 3
- paramValuePairs** = {ArrayList@6377} size = 4
 - 0** = {BasicNameValuePair@6382} "action=postActivation"
 - 1** = {BasicNameValuePair@6383} "sequenceNumber=7"
 - 2** = {BasicNameValuePair@6384} "serialNumber=VDS1000120"
 - 3** = {BasicNameValuePair@6385} "otp=1487006"
- postResponse** = {BasicHttpResponse@6378}
- responseBody** = "message=You have been successfully authenticated.;retCode=0;"
- response** = {HashMap@6390} size = 2
 - 0** = {HashMap\$Node@6398} "message" -> "You have been successfully authenticated."
 - 1** = {HashMap\$Node@6399} "retCode" -> "0"

5.6. Notification Registration

- This service is for Digipass APP to register its push notification identifier (either Google, Apple or any other push service provider) to AccessMatrix UAS server.
- The objective is to store the push notification identifier of the Digipass APP in AccessMatrix UAS server for future use when the application needs to send a push notification message.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "notificationRegistration".	Yes
serialNumber	String	DPMobileGateway expects this parameter as a token serial number.	Yes
sequenceNumber	String	Sequence number of the activated instance (obtain through Digipass SDK) Value: 1,2 3, ,99	Yes
vascoNotificationIdentifier	String	OneSpan push notification identifier generated by OneSpan NotificationClientSDK. (It is not mandatory to use OneSpan NotificationClientSDK.)	Yes

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	A string value of "Notification identifier well registered." will be returned if the push notification identifier registration is successful.

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httpPost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action",
"notificationRegistration"));
    nameValuePairs.add(new BasicNameValuePair("serialNumber", "VDS1000120"));
    nameValuePairs.add(new BasicNameValuePair("sequenceNumber", "1"));
    nameValuePairs.add(new BasicNameValuePair("vascoNotificationIdentifier ",
"123456780987654321abcdefghijkl"));
    httpPost.setEntity(new UrlEncodedFormEntity(nameValuePairs));

    // Execute HTTP Post Request
    HttpResponse postResponse = httpClient.execute(httpPost);
    String responseBody = EntityUtils.toString(postResponse.getEntity());
    response = parseSerializedData(responseBody);

    String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
    String message = response.get("message"); //return message from server.
e.g. success/error message

    /**
     * Continue processing the above returned values
     * with VASCO client-side SDK (e.g. DigipassSDK)
     */
} catch (ClientProtocolException e) {
    // TODO Auto-generated catch block
} catch (IOException e) {
    // TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
    Map<String, String> response = new HashMap<String, String>();
    String[] keyValues = data.split(";");
```

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
for (String keyValue : keyValues)
{
    String[] tmp = keyValue.split("=");
    response.put(tmp[0], tmp[1]);
}
return response;
}
```

5.7. Time Synchronization

- This service is for the Digipass APP to request server time from the AccessMatrix UAS server for time synchronization purposes.
- The server time will be returned in a specific XML format.

Request Method: GET/POST

Request URL: https://<server_host>:<server_port>/<dpmgateway>/DPMobileGateway



Note: <dpmgateway> is the context. The context depends on the name of the WAR file.

Request Parameters

Parameter	Data Type	Description	Mandatory
action	String	This parameter refers to the web service to trigger. In this case, it should be "timeSynchronization".	Yes

Response Result

Parameter	Data Type	Description
retCode	String	DPMobileGateway return code. "0" means the web service request is successful.
message	String	A string value of "Operation Successful." will be returned if the server time is returned successfully.
serverTime	String	A string representation of server time in seconds(milliseconds/1000).

Sample Code

Android

Java Sample for Android sending HTTP POST request to DPMobileGateway

Java

Java Sample for Android sending HTTP POST request to DPMobileGateway

```
HttpClient httpClient = new DefaultHttpClient();
Map<String, String> response = new HashMap<String, String>();

HttpPost httppost = new
HttpPost("https://<server_host>:<server_port>/dpmgateway/DPMobileGateway");
try {
    // Add your data
    List<NameValuePair> nameValuePairs = new ArrayList<NameValuePair>(2);
    nameValuePairs.add(new BasicNameValuePair("action",
"timeSynchronization"));

    // Execute HTTP Post Request
    HttpResponse postResponse = httpClient.execute(httppost);
    String responseBody = EntityUtils.toString(postResponse.getEntity());
    response = parseSerializedData(responseBody);

    String retCode = response.get("retCode"); //return code, 0 means
successful, failed if otherwise
    String message = response.get("message"); //return message from server.
e.g. success/error message
    String serverTime = response.get("serverTime"); //return server time of
UAS server, in seconds
    /**
     * Continue processing the above returned values
     * with VASCO client-side SDK (e.g. DigipassSDK)
     */

} catch (ClientProtocolException e) {
    // TODO Auto-generated catch block
} catch (IOException e) {
    // TODO Auto-generated catch block
}

/**
 * Parses a serialized data (e.g. key=value;...)
 * @param data Data to parse
 * @return Map containing the parsed data
 */
public static Map<String, String> parseSerializedData(String data)
{
    Map<String, String> response = new HashMap<String, String>();
    String[] keyValues = data.split(";");
    for (String keyValue : keyValues)
    {
        String[] tmp = keyValue.split("=");
        response.put(tmp[0], tmp[1]);
    }
    return response;
}
```
